VMware Site Recovery Manager with NetApp ONTAP 9

NetApp Solutions

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TR-4900: VMware Site Recovery Manager with NetApp ONTAP 9

Chance Bingen, NetApp

ONTAP for vSphere

NetApp ONTAP has been a leading storage solution for VMware vSphere environments since its introduction into the modern datacenter in 2002, and it continues to add innovative capabilities to simplify management while reducing costs. This document introduces the ONTAP solution for VMware Site Recovery Manager (SRM), VMware's industry leading disaster recovery (DR) software, including the latest product information and best practices to streamline deployment, reduce risk, and simplify ongoing management.

Best practices supplement other documents such as guides and compatibility tools. They are developed based on lab testing and extensive field experience by NetApp engineers and customers. In some cases, recommended best practices might not be the right fit for your environment; however, they are generally the simplest solutions that meet the needs of the most customers.

This document is focused on capabilities in recent releases of ONTAP 9 when used in conjunction with supported versions of ONTAP tools for VMware vSphere (which includes the NetApp Storage Replication Adapter [SRA] and VASA Provider [VP]), as well as VMware Site Recovery Manager 8. 4.

Why use ONTAP with SRM?

NetApp data management platforms powered by ONTAP software are some of the most widely adopted storage solutions for SRM. The reasons are plentiful: A secure, high performance, unified protocol (NAS and SAN together) data management platform that provides industry defining storage efficiency, multitenancy, quality of service controls, data protection with space-efficient Snapshot copies and replication with SnapMirror. All leveraging native hybrid multi-cloud integration for the protection of VMware workloads and a plethora of automation and orchestration tools at your fingertips.

When you use SnapMirror for array-based replication, you take advantage of one of ONTAP's most proven and mature technologies. SnapMirror gives you the advantage of secure and highly efficient data transfers, copying only changed file system blocks, not entire VMs or datastores. Even those blocks take advantage of space savings, such as deduplication, compression, and compaction. Modern ONTAP systems now use version-independent SnapMirror, allowing you flexibility in selecting your source and destination clusters. SnapMirror has truly become one of the most powerful tools available for disaster recovery.

Whether you are using traditional NFS, iSCSI, or Fibre Channel- attached datastores (now with support for vVols datastores), SRM provides a robust first party offering that leverages the best of ONTAP capabilities for disaster recovery or datacenter migration planning and orchestration.

How SRM leverages ONTAP 9

SRM leverages the advanced data management technologies of ONTAP systems by integrating with ONTAP tools for VMware vSphere, a virtual appliance that includes three primary components:

- The vCenter plug-in, formerly known as Virtual Storage Console (VSC), simplifies storage management and efficiency features, enhances availability, and reduces storage costs and operational overhead, whether you are using SAN or NAS. It uses best practices for provisioning datastores and optimizes ESXi host settings for NFS and block storage environments. For all these benefits, NetApp recommends this

plug-in when using vSphere with systems running ONTAP software.

- The VASA Provider for ONTAP supports the VMware vStorage APIs for Storage Awareness (VASA) framework. VASA Provider connects vCenter Server with ONTAP to aid in provisioning and monitoring VM storage. It enables VMware Virtual Volumes (vVols) support and the management of storage capability profiles (including vVols replication capabilities) and individual VM vVols performance. It also provides alarms for monitoring capacity and compliance with the profiles. When used in conjunction with SRM, the VASA Provider for ONTAP enables support for vVols- based virtual machines without requiring the installation of an SRA adapter on the SRM server.
- The SRA is used together with SRM to manage the replication of VM data between production and disaster recovery sites for traditional VMFS and NFS datastores and also for the nondisruptive testing of DR replicas. It helps automate the tasks of discovery, recovery, and reprotection. It includes both an SRA server appliance and SRA adapters for the Windows SRM server and the SRM appliance.

After you have installed and configured the SRA adapters on the SRM server for protecting non-vVols datastores and/or enabled vVols replication in the VASA Provider settings, you can begin the task of configuring your vSphere environment for disaster recovery.

The SRA and VASA Provider deliver a command-and-control interface for the SRM server to manage the ONTAP FlexVols that contain your VMware Virtual Machines (VMs), as well as the SnapMirror replication protecting them.

Starting with SRM 8.3, a new SRM vVols Provider control path was introduced into the SRM server, allowing it to communicate with the vCenter server and, through it, to the VASA Provider without needing an SRA. This enabled the SRM server to leverage much deeper control over the ONTAP cluster than was possible before, because VASA provides a complete API for closely coupled integration.

SRM can test your DR plan nondisruptively using NetApp's proprietary FlexClone technology to make nearly instantaneous clones of your protected datastores at your DR site. SRM creates a sandbox to safely test so that your organization, and your customers, are protected in the event of a true disaster, giving you confidence in your organizations ability to execute a failover during a disaster.

In the event of a true disaster or even a planned migration, SRM allows you to send any last-minute changes to the dataset via a final SnapMirror update (if you choose to do so). It then breaks the mirror and mounts the datastore to your DR hosts. At that point, your VMs can be automatically powered up in any order according to your pre-planned strategy.

SRM with ONTAP and other use cases: hybrid cloud and migration

Integrating your SRM deployment with ONTAP advanced data management capabilities allows for vastly improved scale and performance when compared with local storage options. But more than that, it brings the flexibility of the hybrid cloud. The hybrid cloud enables you to save money by tiering unused data blocks from your high-performance array to your preferred hyperscaler using FabricPool, which could be an on-premises S3 store such as NetApp StorageGRID. You can also use SnapMirror for edge-based systems with software-defined ONTAP Select or cloud-based DR using Cloud Volumes ONTAP (CVO) or NetApp Private Storage in Equinix for Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) to create a fully integrated storage, networking, and compute- services stack in the cloud.

You could then perform test failover inside a cloud service provider's datacenter with near-zero storage footprint thanks to FlexClone. Protecting your organization can now cost less than ever before.

SRM can also be used to execute planned migrations by leveraging SnapMirror to efficiently transfer your VMs from one datacenter to another or even within the same datacenter, whether your own, or via any number of

NetApp partner service providers.

New features with SRM and ONTAP Tools

With the transition from the legacy virtual appliance, ONTAP tools brings a wealth of new features, higher limits, and new vVols support.

Latest versions of vSphere and Site Recovery Manager

With the release of SRM 8.3 and later and the 9.7.1 and later releases of ONTAP tools, you are now able to protect VMs running on VMware vSphere 7.

NetApp has shared a deep partnership with VMware for nearly two decades and strives to provide support for the latest releases as soon as possible. Always check the NetApp Interoperability Matrix Tool (IMT) for the latest qualified combinations of software.

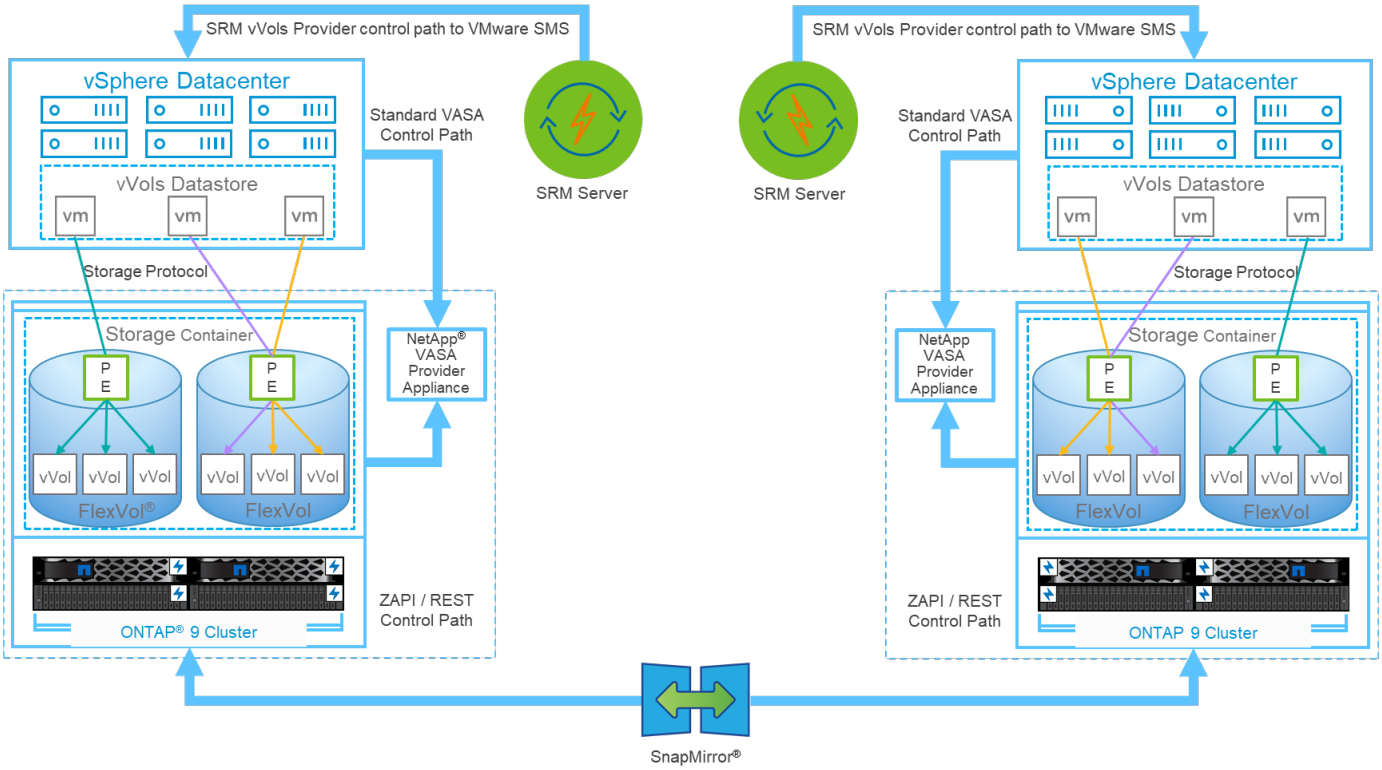
The NetApp IMT can be found here.

vVols support (and why SPBM matters, even with SRM)

Starting with the 8.3 release, SRM now supports storage policy-based management (SPBM) of replication leveraging vVols and array-based replication. To accomplish this, the SRM server was updated to include a new SRM vVols provider service, which communicates to the vCenter server's SMS service for VASA related tasks.

One advantage to this architecture is that an SRA is no longer needed since everything is handled using VASA.

SPBM is a powerful tool in the vSphere toolbox, allow simplified, predictable, and consistent storage services for consumption by automation frameworks in private and hybrid cloud environments. Fundamentally, SPBM allows you to define classes of service that meet the needs of your diverse customer base. SRM now allows you to expose replication capabilities to your customers for critical workloads requiring robust industry-standard disaster- recovery orchestration and automation.



vVols Architecture 2.3 Support for appliance-based SRM servers

Photon OS-based SRM servers are now supported, in addition to legacy Windows-based platforms.

You can now install SRA adapters regardless of your preferred SRM server type.

Support for IPv6

IPv6 is now supported with the following limitations:

- vCenter 6.7 or later
- Not supported with SRM 8.2 (8.1, 8.3, and 8. 4 are supported)
- Check the Interoperability Matrix Tool for the latest qualified versions.

Improved performance

Operational performance is a key requirement for SRM task execution. To meet the requirements of modern RTOs and RPOs, the SRA with ONTAP tools has added two new improvements.

- **Support for concurrent reprotect operations.** First introduced in SRA 9.7.1, enabling this feature allows you to run reprotect on two or more recovery plans concurrently, thus reducing the time required to reprotect datastores after a failover or migration and remain within your RTO and RPO parameters.
- **ONTAP Tools 9.8 adds a new NAS- only optimized mode.** When you use SVM- scoped accounts and connections to ONTAP clusters with only NFS based datastores, you can enable NAS-only optimized mode for peak performance in supported environments.

Greater scale

The ONTAP tools SRA can now support up to 500 protection groups (PGs) when used with SRM 8.3 and later.

Synchronous replication

A long awaited and much anticipated new feature is SnapMirror Synchronous (SM-S) with ONTAP 9.5 and later which delivers a volume granular zero RPO data replication solution for your mission-critical applications. SM-S requires ONTAP tools 9.8 or later.

REST API support

SRA server configuration can now be managed by REST APIs. A Swagger UI has been added to assist in building your automation workflows and can be found on your ONTAP tools appliance at `https://<appliance>:8143/api/rest/swagger-ui.html#/`.

Deployment best practices

SVM layout and segmentation for SMT

With ONTAP, the concept of the storage virtual machine (SVM) provides strict segmentation in secure multitenant environments. SVM users on one SVM cannot access or manage resources from another. In this way, you can leverage ONTAP technology by creating separate SVMs for different business units who manage their own SRM workflows on the same cluster for greater overall storage efficiency.

Consider managing ONTAP using SVM-scoped accounts and SVM management LIFs to not only improve security controls, but also improve performance. Performance is inherently greater when using SVM-scoped connections because the SRA is not required to process all the resources in an entire cluster, including physical resources. Instead, it only needs to understand the logical assets that are abstracted to the particular SVM.

When using NAS protocols only (no SAN access), you can even leverage the new NAS optimized mode by setting the following parameter (note that the name is such because SRA and VASA use the same backend services in the appliance):

1. Log into the control panel at `https://<IP address>:9083` and click Web based CLI interface.
2. Run the command `vp updateconfig -key=enable.qtree.discovery -value=true`.
3. Run the command `vp updateconfig -key=enable.optimised.sra -value=true`.
4. Run the command `vp reloadconfig`.

Deploy ONTAP tools and considerations for vVols

If you intend to use SRM with vVols, you must manage the storage using cluster- scoped credentials and a cluster management LIF. This is because the VASA Provider must understand the underlying physical architecture to satisfy the policy requires for VM storage policies. For example, if you have a policy that requires all- flash storage, the VASA Provider must be able to see which systems are all flash.

Another deployment best practice is to never store your ONTAP tools appliance on a vVols datastore that it is managing. This could lead to a situation whereby you cannot power on the VASA Provider because you cannot create the swap vVol for the appliance because the appliance is offline.

Best practices for managing ONTAP 9 systems

As previously mentioned, you can manage ONTAP clusters using either cluster or SVM scoped credentials and management LIFs. For optimum performance, you may want to consider using SVM- scoped credentials whenever you aren't using vVols. However, in doing so, you should be aware of some requirements, and that you do lose some functionality.

- The default vsadmin SVM account does not have the required access level to perform ONTAP tools tasks. Therefore, you need to create a new SVM account.
- If you are using ONTAP 9.8 or later, NetApp recommends creating an RBAC least privileged user account using ONTAP System Manager's users menu together with the JSON file available on your ONTAP tools appliance at `https://<IP address>:9083/vsc/config/`. Use your administrator password to download the JSON file. This can be used for SVM or cluster scoped accounts.

  If you are using ONTAP 9.6 or earlier, you should use the RBAC User Creator (RUC) tool available in the NetApp Support Site Toolchest.

- Because the vCenter UI plugin, VASA Provider, and SRA server are all fully integrated services, you must add storage to the SRA adapter in SRM the same way you add storage in the vCenter UI for ONTAP tools. Otherwise, the SRA server might not recognize the requests being sent from SRM via the SRA adapter.
- NFS path checking is not performed when using SVM-scoped credentials. This is because the physical location is logically abstracted from the SVM. This is not a cause for concern though, as modern ONTAP systems no longer suffer any noticeable performance decline when using indirect paths.
- Aggregate space savings due to storage efficiency might not be reported.
- Where supported, load-sharing mirrors cannot be updated.
- EMS logging might not be performed on ONTAP systems managed with SVM scoped credentials.

Operational best practices

Datastores and protocols

If possible, always use ONTAP tools to provision datastores and volumes. This makes sure that volumes, junction paths, LUNs, igroups, export policies, and other settings are configured in a compatible manner.

SRM supports iSCSI, Fibre Channel, and NFS version 3 with ONTAP 9 when using array-based replication through SRA. SRM does not support array-based replication for NFS version 4.1 with either traditional or vVols datastores.

To confirm connectivity, always verify that you can mount and unmount a new test datastore at the DR site from the destination ONTAP cluster. Test each protocol you intend to use for datastore connectivity. A best practice is to use ONTAP tools to create your test datastore, since it is doing all the datastore automation as directed by SRM.

SAN protocols should be homogeneous for each site. You can mix NFS and SAN, but the SAN protocols should not be mixed within a site. For example, you can use FCP in site A, and iSCSI in site B. You should not use both FCP and iSCSI at site A. The reason for this is that the SRA does not create mixed igroups at the recovery site and SRM does not filter the initiator list given to the SRA.

Previous guides advised to create LIF to data locality. That is to say, always mount a datastore using a LIF located on the node that physically owns the volume. That is no longer a requirement in modern versions of ONTAP 9. Whenever possible, and if given cluster scoped credentials, ONTAP tools will still choose to load balance across LIFs local to the data, but it is not a requirement for high availability or performance.

NetApp ONTAP 9 can be configured to automatically remove Snapshot copies to preserve uptime in the event of an out-of-space condition when autosize is not able to supply sufficient emergency capacity. The default setting for this capability does not automatically delete the Snapshot copies that are created by SnapMirror. If SnapMirror Snapshot copies are deleted, then the NetApp SRA cannot reverse and resynchronize replication for the affected volume. To prevent ONTAP from deleting SnapMirror Snapshot copies, configure the Snapshot autodelete capability to try.

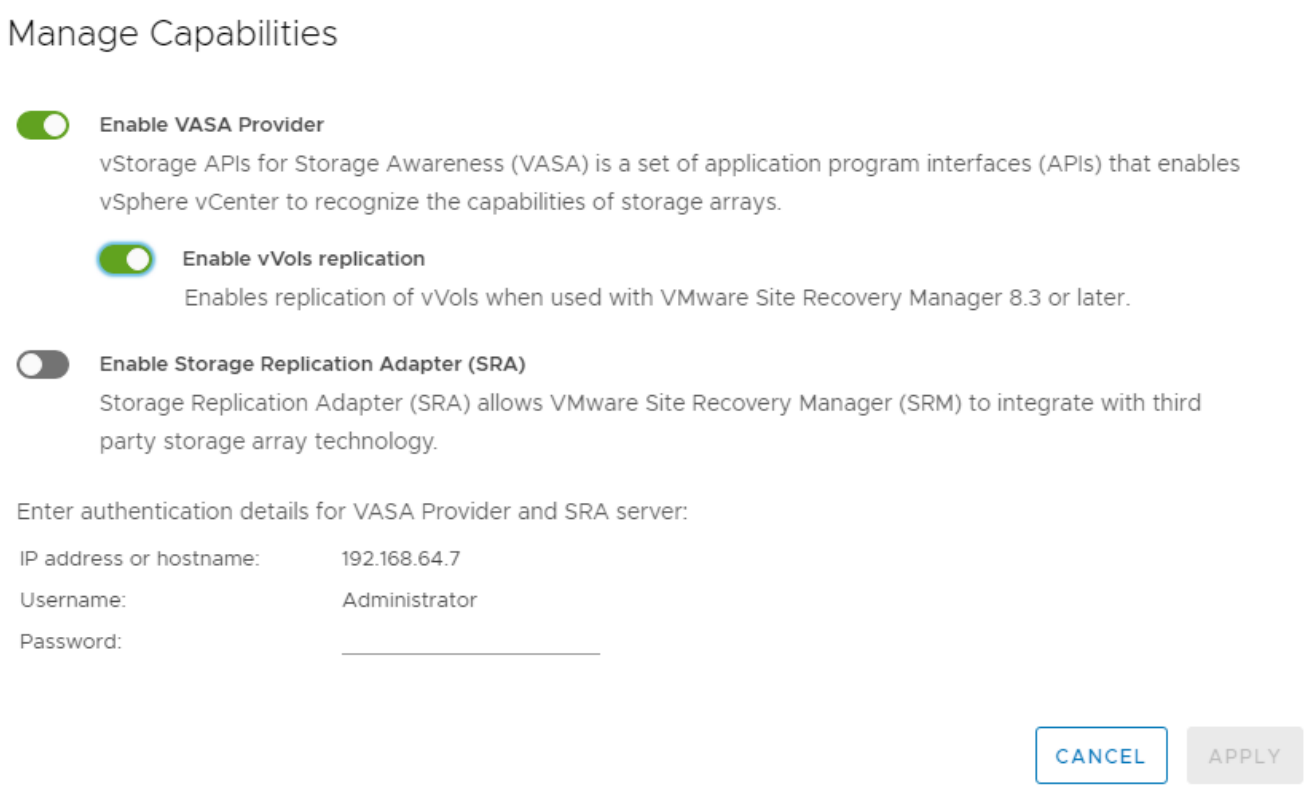
```
snap autodelete modify -volume -commitment try
```

Volume autosize should be set to `grow` for volumes containing SAN datastores and `grow_shrink` for NFS datastores. Refer to the ONTAP 9 Documentation Center for specific syntax.

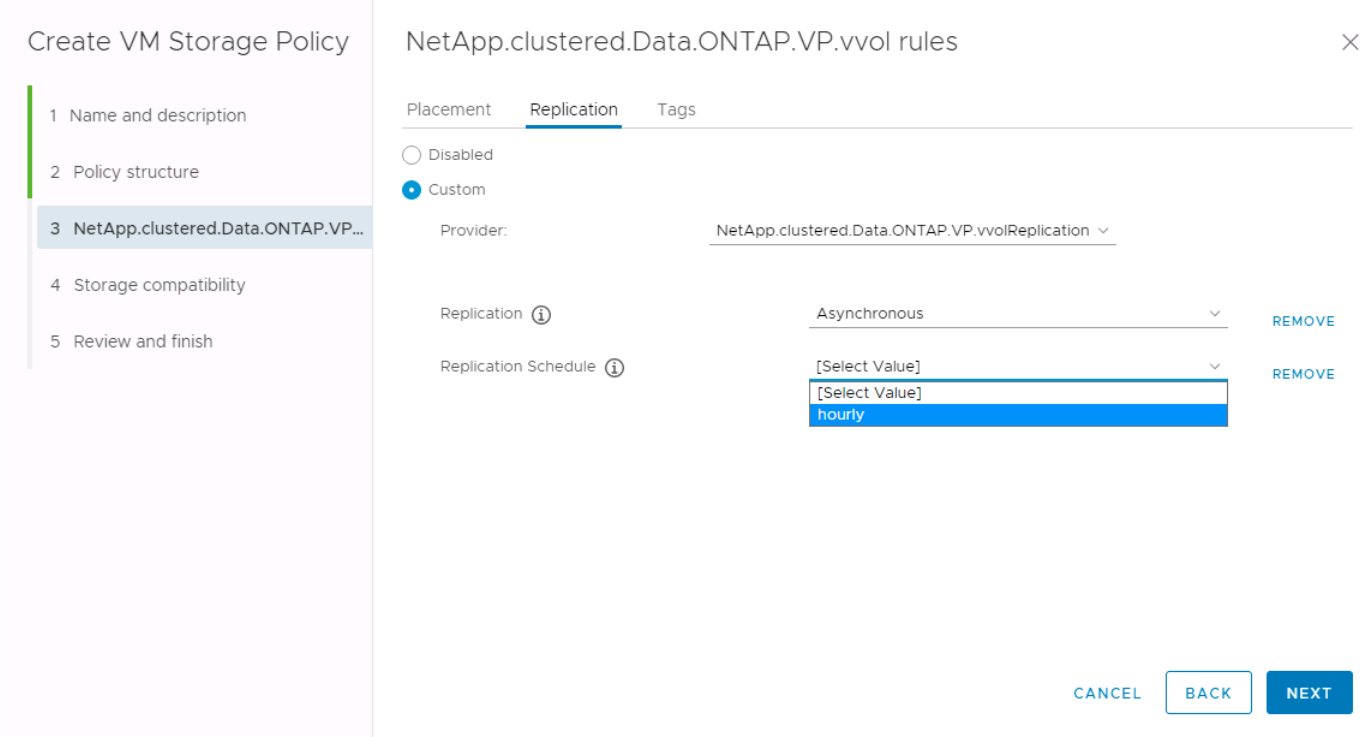
SPBM and vVols

Starting with SRM 8.3, protection of VMs using vVols datastores is supported. SnapMirror schedules are exposed to VM storage policies by the VASA Provider when vVols replication is enabled in the ONTAP tools settings menu, as shown in the following screenshots.

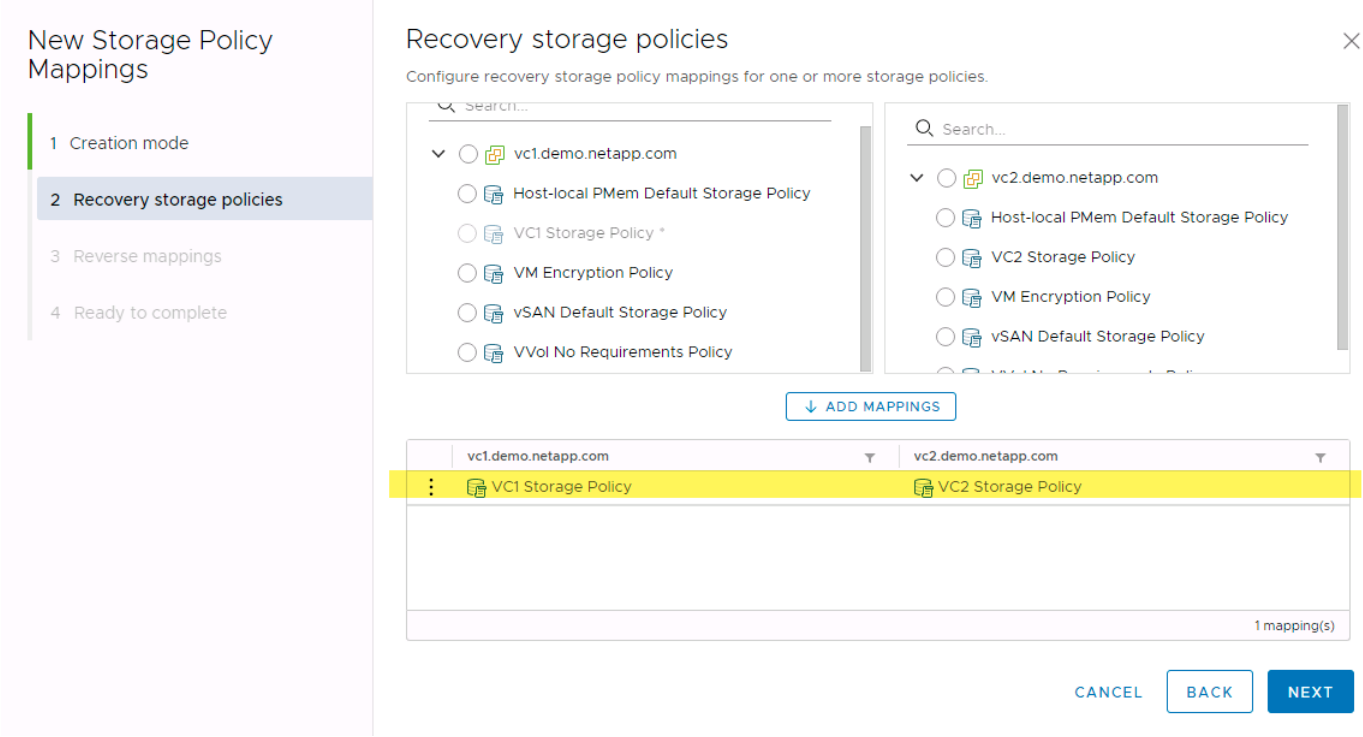
The following example show the enablement of vVols replication.



The following screenshot provides an example of SnapMirror schedules displayed in the Create VM Storage Policy wizard.



The ONTAP VASA Provider supports failover to dissimilar storage. For example, the system can fail over from ONTAP Select at an edge location to an AFF system in the core datacenter. Regardless of storage similarity, you must always configure storage policy mappings and reverse mappings for replication-enabled VM storage policies to make sure that services provided at the recovery site meet expectations and requirements. The following screenshot highlights a sample policy mapping.

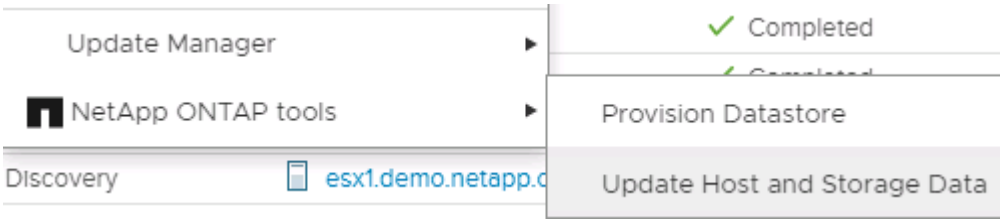


Create replicated volumes for vVols datastores

Unlike previous vVols datastores, replicated vVols datastores must be created from the start with replication

enabled, and they must use volumes that were pre-created on the ONTAP systems with SnapMirror relationships. This requires pre-configuring things like cluster peering and SVM peering. These activities should be performed by your ONTAP administrator, because this facilitates a strict separation of responsibilities between those who manage the ONTAP systems across multiple sites and those who are primarily responsible for vSphere operations.

This does come with a new requirement on behalf of the vSphere administrator. Because volumes are being created outside the scope of ONTAP tools, it is unaware of the changes your ONTAP administrator has made until the regularly scheduled rediscovery period. For that reason, it is a best practice to always run rediscovery whenever you create a volume or SnapMirror relationship to be used with vVols. Simply right click on the host or cluster and select NetApp ONTAP tools > Update Host and Storage Data, as shown in the following screenshot.

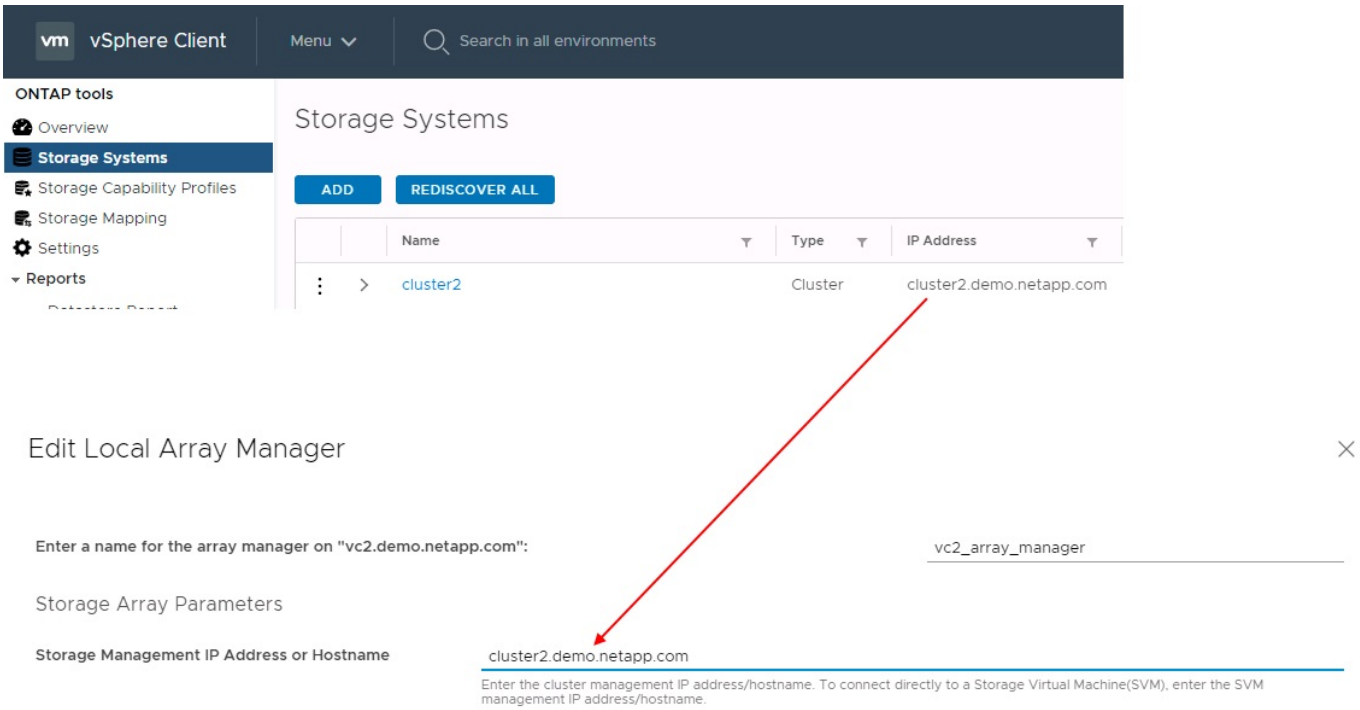


One caution should be taken when it comes to vVols and SRM. Never mix protected and unprotected VMs in the same vVols datastore. The reason for this is that when you use SRM to failover to your DR site, only those VMs that are part of the protection group are brought online in DR. Therefore, when you reprotect (reverse the SnapMirror from DR back to production again), you may overwrite the VMs that were not failed over and could contain valuable data.

About array pairs

An array manager is created for each array pair. With SRM and ONTAP tools, each array pairing is done with the scope of an SVM, even if you are using cluster credentials. This allows you to segment DR workflows between tenants based on which SVMs they have been assigned to manage. You can create multiple array managers for a given cluster, and they can be asymmetric in nature. You can fan out or fan in between different ONTAP 9 clusters. For example, you can have SVM-A and SVM-B on Cluster-1 replicating to SVM-C on Cluster-2, SVM-D on Cluster-3, or vice-versa.

When configuring array pairs in SRM, you should always add them in SRM the same way as you added them to ONTAP Tools, meaning, they must use the same username, password, and management LIF. This requirement ensures that SRA communicates properly with the array. The following screenshot illustrates how a cluster might appear in ONTAP Tools and how it might be added to an array manager.



About replication groups

Replication groups contain logical collections of virtual machines that are recovered together. The ONTAP tools VASA Provider automatically creates replication groups for you. Because ONTAP SnapMirror replication occurs at the volume level, all VMs in a volume are in the same replication group.

There are several factors to consider with replication groups and how you distribute VMs across FlexVol volumes. Grouping similar VMs in the same volume can increase storage efficiency with older ONTAP systems that lack aggregate- level deduplication, but grouping increases the size of the volume and reduces volume I/O concurrency. The best balance of performance and storage efficiency can be achieved in modern ONTAP systems by distributing VMs across FlexVol volumes in the same aggregate, thereby leveraging aggregate level deduplication and gaining greater I/O parallelization across multiple volumes. You can recover VMs in the volumes together because a protection group (discussed below) can contain multiple replication groups. The downside to this layout is that blocks might be transmitted over the wire multiple times because volume SnapMirror doesn't take aggregate deduplication into account.

One final consideration for replication groups is that each one is by its nature a logical consistency group (not to be confused with SRM consistency groups). This is because all VMs in the volume are transferred together using the same snapshot. So if you have VMs that must be consistent with each other, consider storing them in the same FlexVol.

About protection groups

Protection groups define VMs and datastores in groups that are recovered together from the protected site. The protected site is where the VMs that are configured in a protection group exist during normal steady-state operations. It is important to note that even though SRM might display multiple array managers for a protection group, a protection group cannot span multiple array managers. For this reason, you should not span VM files across datastores on different SVMs.

About recovery plans

Recovery plans define which protection groups are recovered in the same process. Multiple protection groups can be configured in the same recovery plan. Also, to enable more options for the execution of recovery plans, a single protection group can be included in multiple recovery plans.

Recovery plans allow SRM administrators to define recovery workflows by assigning VMs to a priority group from 1 (highest) to 5 (lowest), with 3 (medium) being the default. Within a priority group, VMs can be configured for dependencies.

For example, your company could have a tier-1 business critical application that relies on a Microsoft SQL server for its database. So, you decide to place your VMs in priority group 1. Within priority group 1, you begin planning the order to bring up services. You probably want your Microsoft Windows domain controller to boot up before your Microsoft SQL server, which would need to be online before your application server, and so on. You would add all these VMs to the priority group and then set the dependencies, because dependencies only apply within a given priority group.

NetApp strongly recommends working with your application teams to understand the order of operations required in a failover scenario and to construct your recovery plans accordingly.

Test failover

As a best practice, always perform a test failover whenever a change is made to the configuration of a protected VM storage. This ensures that, in the event of a disaster, you can trust that Site Recovery Manager is able to restore services within the expected RTO target.

NetApp also recommends confirming in-guest application functionality occasionally, especially after reconfiguring VM storage.

When a test recovery operation is performed, a private test bubble network is created on the ESXi host for the VMs. However, this network is not automatically connected to any physical network adapters and therefore does not provide connectivity between the ESXi hosts. To allow communication among VMs that are running on different ESXi hosts during DR testing, a physical private network is created between the ESXi hosts at the DR site. To verify that the test network is private, the test bubble network can be separated physically or by using VLANs or VLAN tagging. This network must be segregated from the production network because as the VMs are recovered, they cannot be placed on the production network with IP addresses that could conflict with actual production systems. When a recovery plan is created in SRM, the test network that was created can be selected as the private network to connect the VMs to during the test.

After the test has been validated and is no longer required, perform a cleanup operation. Running cleanup returns the protected VMs to their initial state and resets the recovery plan to the Ready state.

Failover considerations

There are several other considerations when it comes to failing over a site in addition to the order of operations mentioned in this guide.

One issue you might have to contend with is networking differences between sites. Some environments might be able to use the same network IP addresses at both the primary site and the DR site. This ability is referred to as a stretched virtual LAN (VLAN) or stretched network setup. Other environments might have a requirement to use different network IP addresses (for example, in different VLANs) at the primary site relative to the DR site.

VMware offers several ways to solve this problem. For one, network virtualization technologies like VMware NSX-T Data Center abstract the entire networking stack from layers 2 through 7 from the operating

environment, allowing for more portable solutions. You can read more about NSX-T options with SRM here.

SRM also gives you the ability to change the network configuration of a VM as it is recovered. This reconfiguration includes settings such as IP addresses, gateway address, and DNS server settings. Different network settings, which are applied to individual VMs as they are recovered, can be specified in the property's settings of a VM in the recovery plan.

To configure SRM to apply different network settings to multiple VMs without having to edit the properties of each one in the recovery plan, VMware provides a tool called the dr-ip-customizer. For information on how to use this utility, refer to VMware's documentation here.

Reprotect

After a recovery, the recovery site becomes the new production site. Because the recovery operation broke the SnapMirror replication, the new production site is not protected from any future disaster. A best practice is to protect the new production site to another site immediately after a recovery. If the original production site is operational, the VMware administrator can use the original production site as a new recovery site to protect the new production site, effectively reversing the direction of protection. Reprotection is available only in non-catastrophic failures. Therefore, the original vCenter Servers, ESXi servers, SRM servers, and corresponding databases must be eventually recoverable. If they are not available, a new protection group and a new recovery plan must be created.

Failback

A failback operation is fundamentally a failover in a different direction than before. As a best practice, you verify that the original site is back to acceptable levels of functionality before attempting to failback, or, in other words, failover to the original site. If the original site is still compromised, you should delay failback until the failure is sufficiently remediated.

Another failback best practice is to always perform a test failover after completing reprotect and before doing your final failback. This verifies that the systems in place at the original site can complete the operation.

Reprotecting the original site

After failback, you should confirm with all stake holders that their services have been returned to normal before running reprotect again,

Running reprotect after failback essentially puts the environment back in the state it was in at the beginning, with SnapMirror replication again running from the production site to the recovery site.

Replication topologies

In ONTAP 9, the physical components of a cluster are visible to cluster administrators, but they are not directly visible to the applications and hosts that use the cluster. The physical components provide a pool of shared resources from which the logical cluster resources are constructed. Applications and hosts access data only through SVMs that contain volumes and LIFs.

Each NetApp SVM is treated as an array in VMware vCenter Site Recovery Manager. SRM supports certain array-to-array (or SVM-to-SVM) replication layouts.

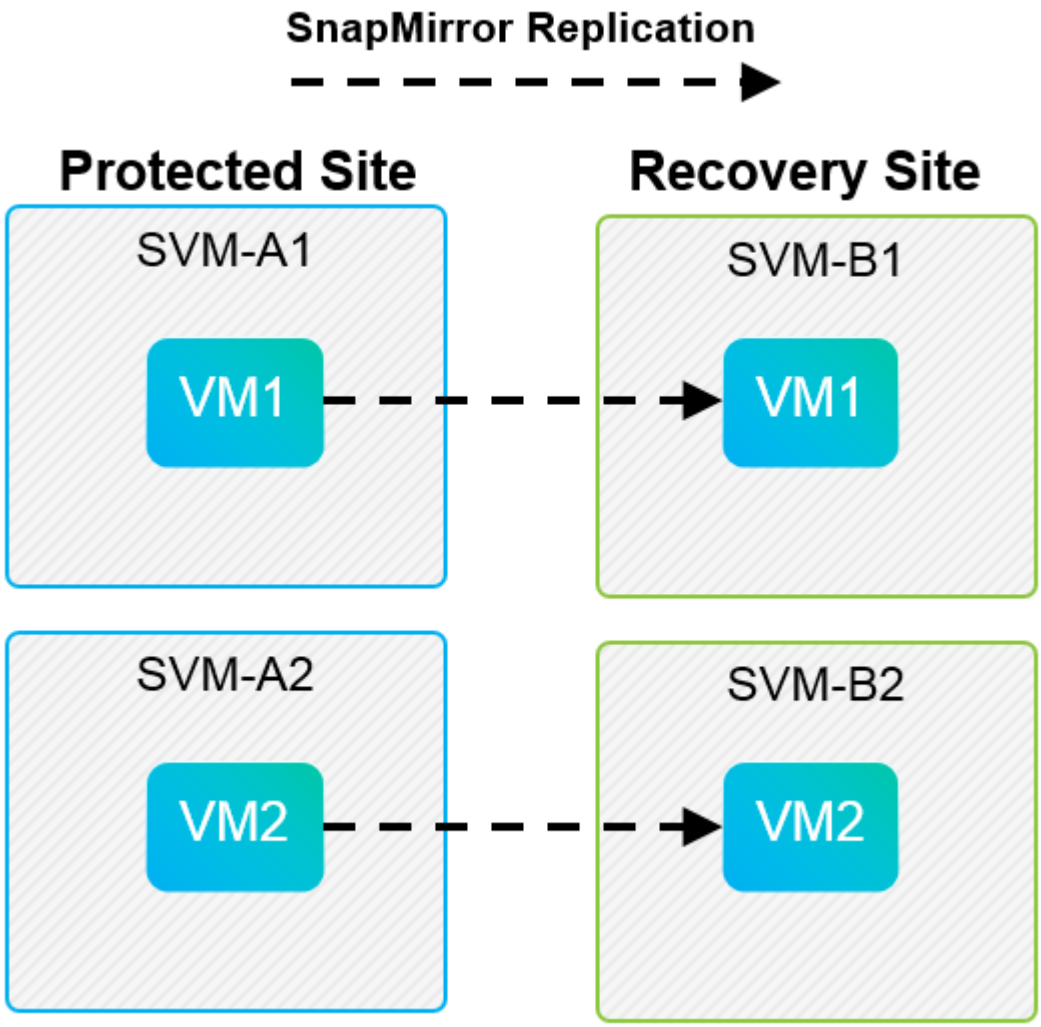
A single VM cannot own data—Virtual Machine Disk (VMDK) or RDM—on more than one SRM array for the following reasons:

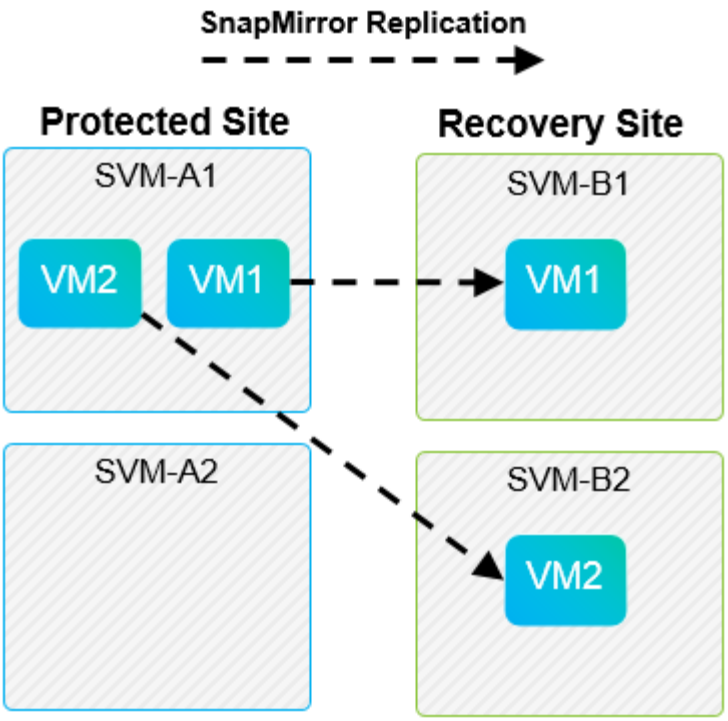
- SRM sees only the SVM, not an individual physical controller.
- An SVM can control LUNs and volumes that span multiple nodes in a cluster.

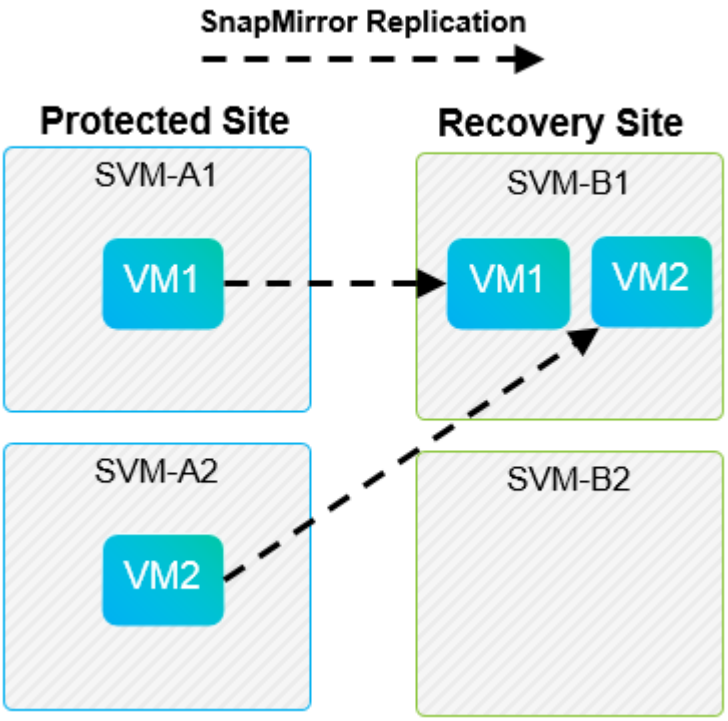
Best Practice
To determine supportability, keep this rule in mind: to protect a VM by using SRM and the NetApp SRA, all parts of the VM must exist on only one SVM. This rule applies at both the protected site and the recovery site.

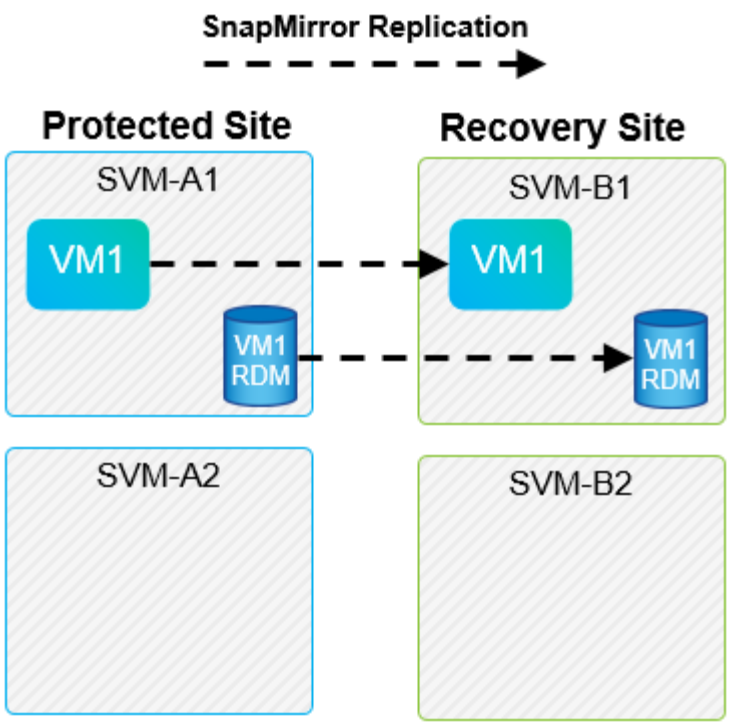
Supported SnapMirror layouts

The following figures show the SnapMirror relationship layout scenarios that SRM and SRA support. Each VM in the replicated volumes owns data on only one SRM array (SVM) at each site.



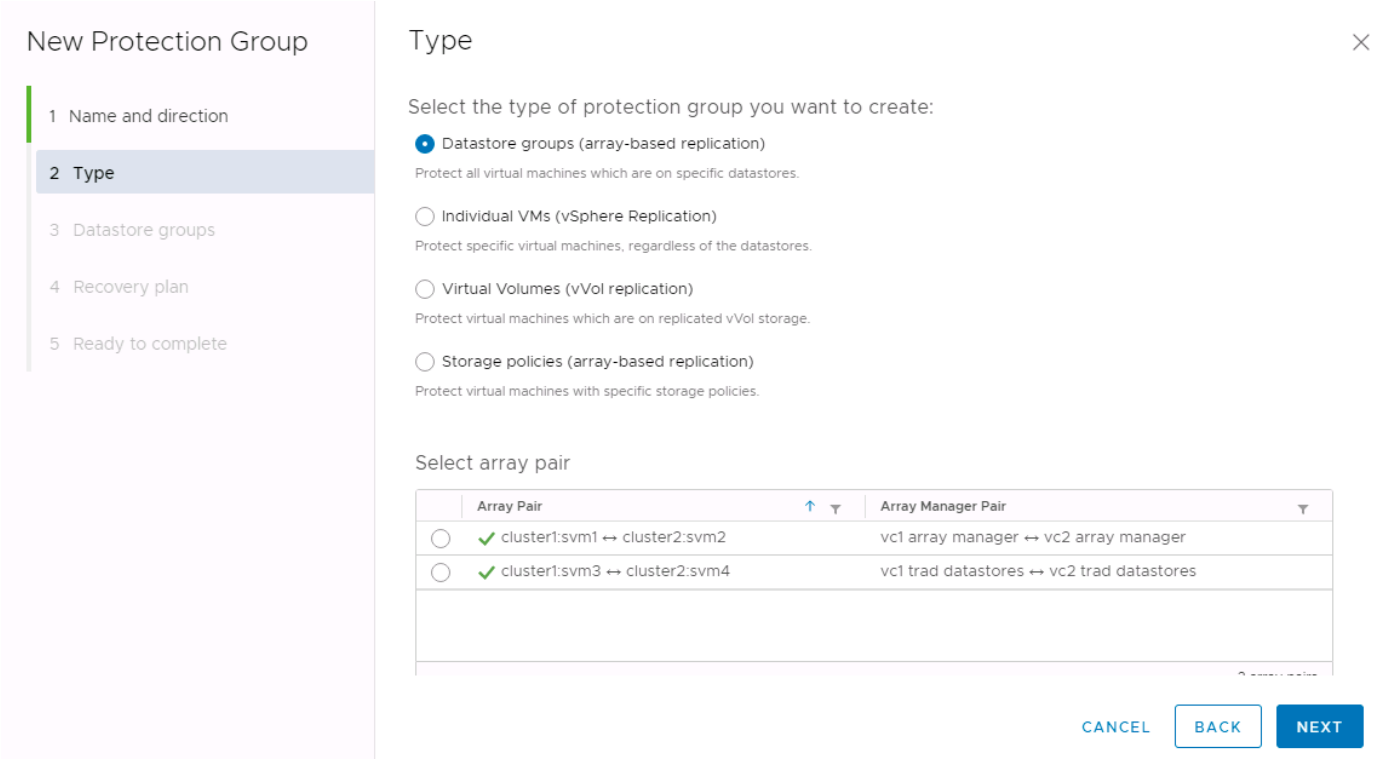

SnapMirror Replication
Protected Site
Recovery Site
SVM-A1
VM2
VM1
SVM-B1
VM1
SVM-A2
SVM-B2
VM2


SnapMirror Replication
Protected Site
Recovery Site
SVM-A1
VM1
SVM-B1
VM1
VM2
SVM-A2
VM2
SVM-B2



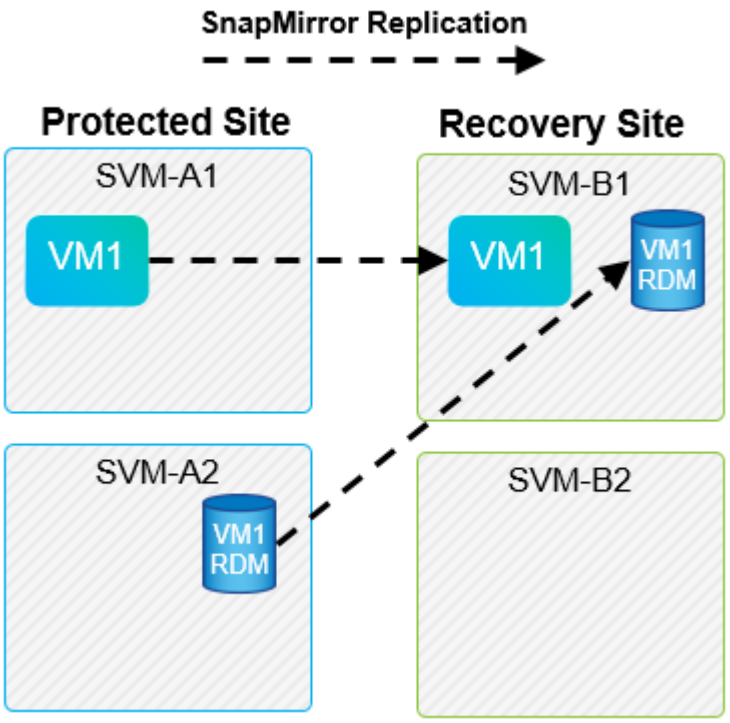
Supported Array Manager layouts

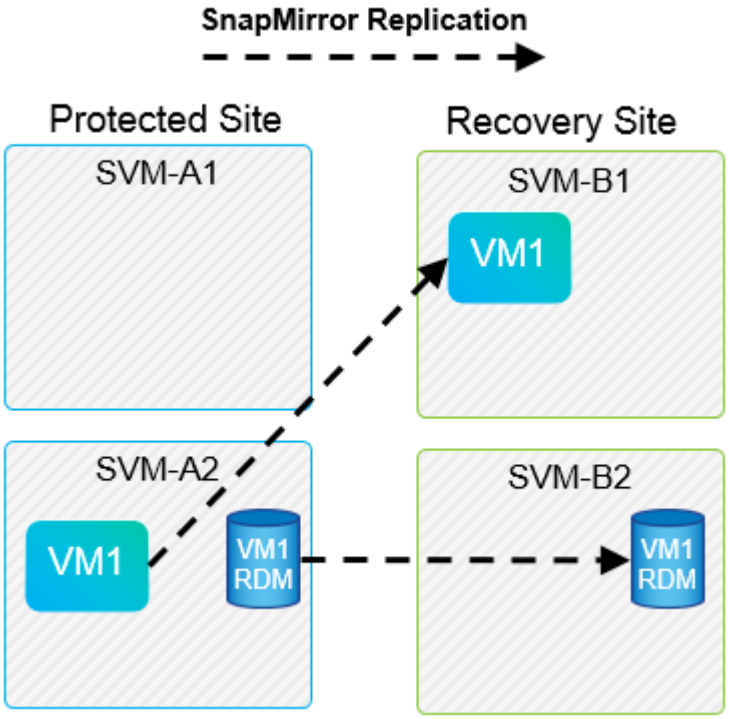
When you use array-based replication (ABR) in SRM, protection groups are isolated to a single array pair, as shown in the following screenshot. In this scenario, `SVM1` and `SVM2` are peered with `SVM3` and `SVM4` at the recovery site. However, you can select only one of the two array pairs when you create a protection group.



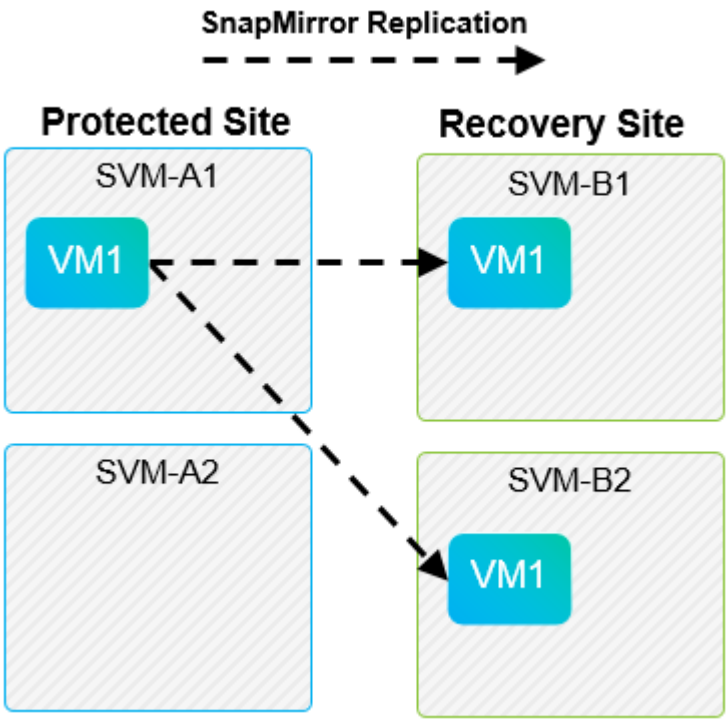
Unsupported layouts

Unsupported configurations have data (VMDK or RDM) on multiple SVMs that is owned by an individual VM. In the examples shown in the following figures, `VM1` cannot be configured for protection with SRM because `VM1` has data on two SVMs.



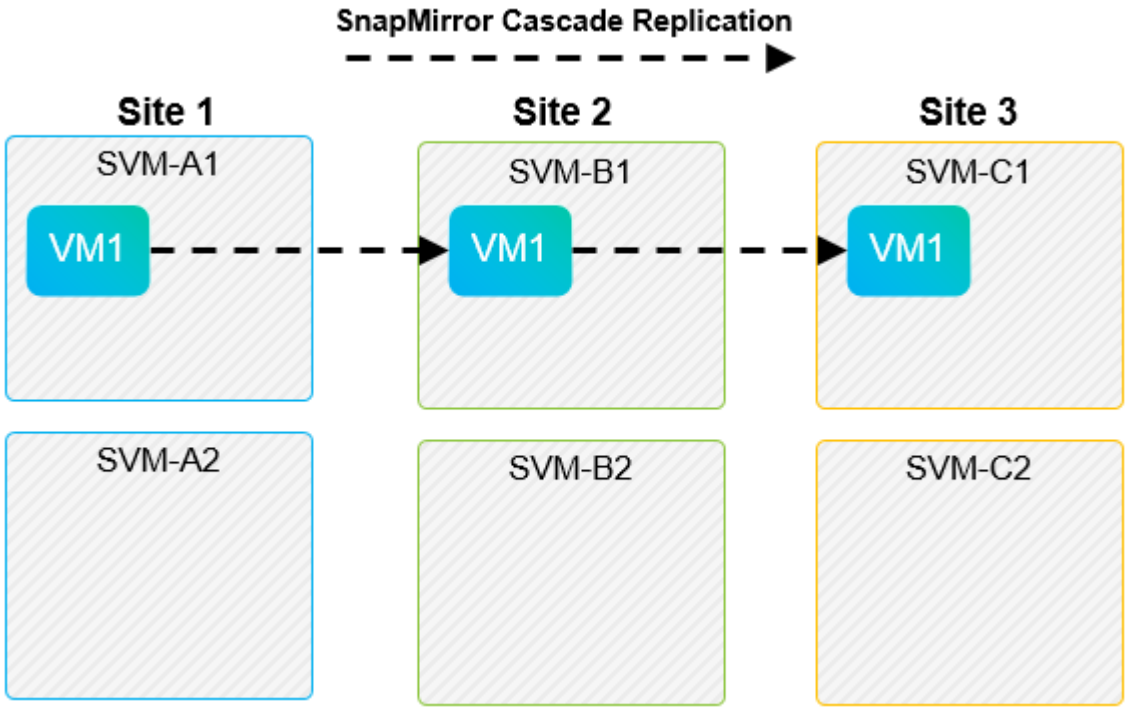


Any replication relationship in which an individual NetApp volume is replicated from one source SVM to multiple destinations in the same SVM or in different SVMs is referred to as SnapMirror fan-out. Fan-out is not supported with SRM. In the example shown in the following figure, `VM1` cannot be configured for protection in SRM because it is replicated with SnapMirror to two different locations.



SnapMirror cascade

SRM does not support cascading of SnapMirror relationships, in which a source volume is replicated to a destination volume and that destination volume is also replicated with SnapMirror to another destination volume. In the scenario shown in the following figure, SRM cannot be used for failover between any sites.



SnapMirror and SnapVault

NetApp SnapVault software enables disk-based backup of enterprise data between NetApp storage systems. SnapVault and SnapMirror can coexist in the same environment; however, SRM supports the failover of only

the SnapMirror relationships.

> The NetApp SRA supports the `mirror-vault` policy type.

SnapVault was rebuilt from the ground up for ONTAP 8.2. Although former Data ONTAP 7-Mode users should find similarities, major enhancements have been made in this version of SnapVault. One major advance is the ability to preserve storage efficiencies on primary data during SnapVault transfers.

An important architectural change is that SnapVault in ONTAP 9 replicates at the volume level as opposed to at the qtree level, as is the case in 7-Mode SnapVault. This setup means that the source of a SnapVault relationship must be a volume, and that volume must replicate to its own volume on the SnapVault secondary system.

In an environment in which SnapVault is used, specifically named Snapshot copies are created on the primary storage system. Depending on the configuration implemented, the named Snapshot copies can be created on the primary system by a SnapVault schedule or by an application such as NetApp Active IQ Unified Manager. The named Snapshot copies that are created on the primary system are then replicated to the SnapMirror destination, and from there they are vaulted to the SnapVault destination.

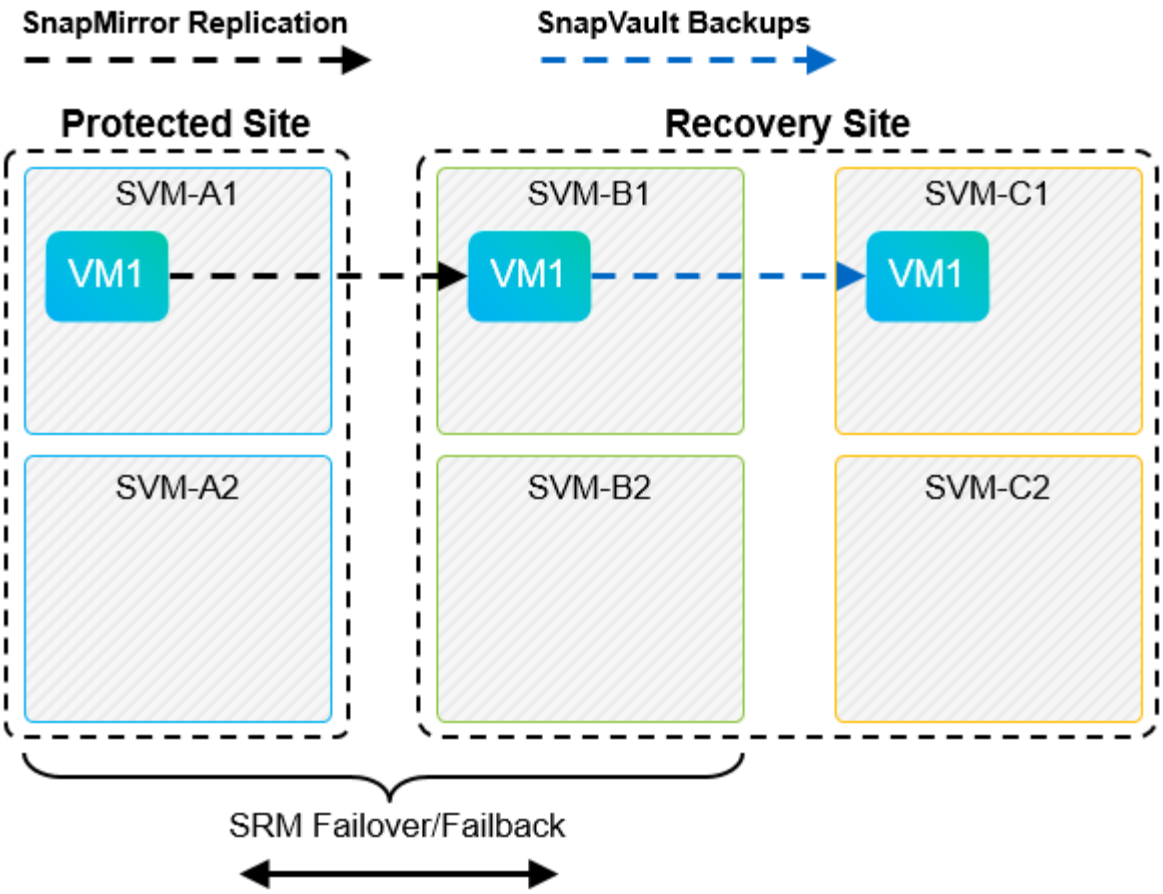
A source volume can be created in a cascade configuration in which a volume is replicated to a SnapMirror destination in the DR site, and from there it is vaulted to a SnapVault destination. A source volume can also be created in a fan-out relationship in which one destination is a SnapMirror destination and the other destination is a SnapVault destination. However, SRA does not automatically reconfigure the SnapVault relationship to use the SnapMirror destination volume as the source for the vault when SRM failover or replication reversal occurs.

For the latest information about SnapMirror and SnapVault for ONTAP 9, see TR-4015 SnapMirror Configuration Best Practice Guide for ONTAP 9.

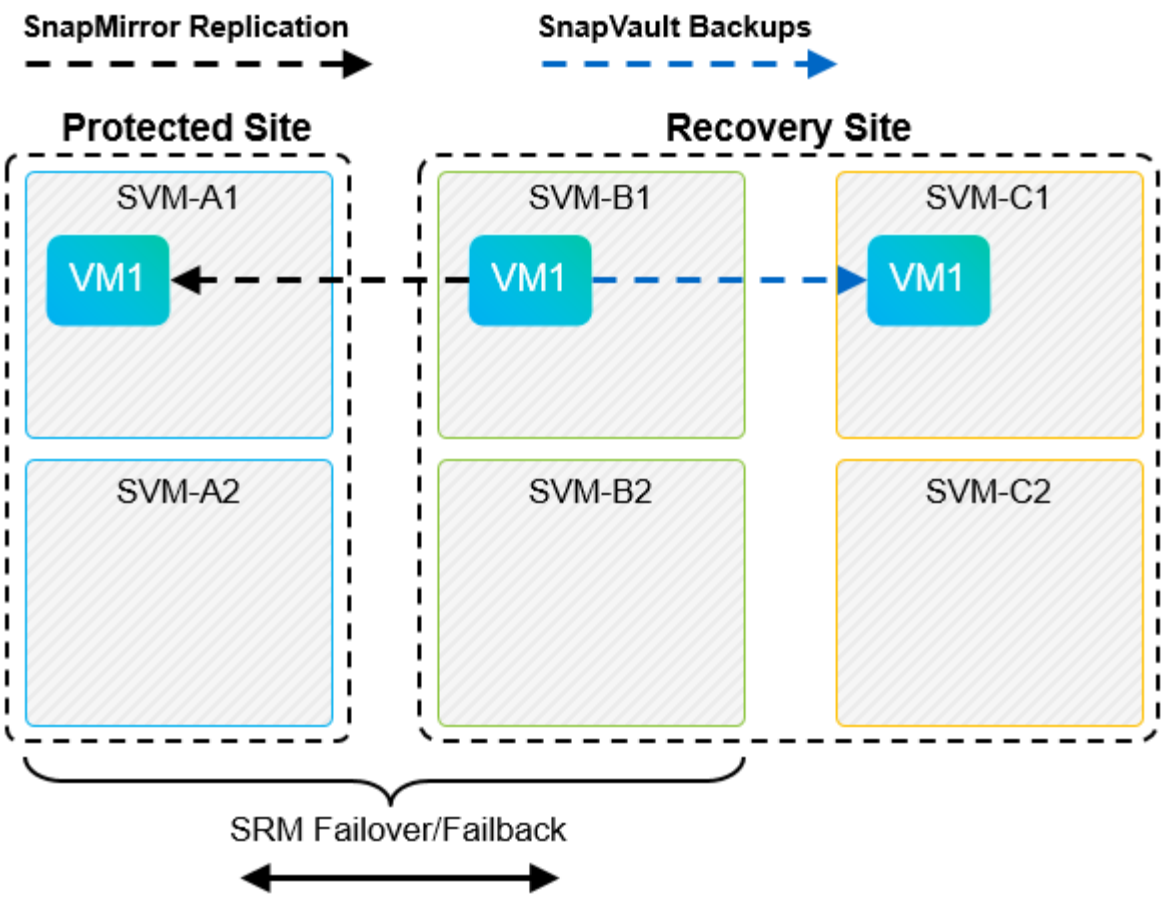
Best Practice
If SnapVault and SRM are used in the same environment, NetApp recommends using a SnapMirror to SnapVault cascade configuration in which SnapVault backups are normally performed from the SnapMirror destination at the DR site. In the event of a disaster, this configuration makes the primary site inaccessible. Keeping the SnapVault destination at the recovery site allows SnapVault backups to be reconfigured after failover so that SnapVault backups can continue while operating at the recovery site.

In a VMware environment, each datastore has a universal unique identifier (UUID), and each VM has a unique managed object ID (MOID). These IDs are not maintained by SRM during failover or failback. Because datastore UUIDs and VM MOIDs are not maintained during failover by SRM, any applications that depend on these IDs must be reconfigured after SRM failover. An example application is NetApp Active IQ Unified Manager, which coordinates SnapVault replication with the vSphere environment.

The following figure depicts a SnapMirror to SnapVault cascade configuration. If the SnapVault destination is at the DR site or at a tertiary site that is not affected by an outage at the primary site, the environment can be reconfigured to allow backups to continue after failover.



The following figure depicts the configuration after SRM has been used to reverse SnapMirror replication back to the primary site. The environment has also been reconfigured such that SnapVault backups are occurring from what is now the SnapMirror source. This setup is a SnapMirror SnapVault fan-out configuration.



After SRM performs failback and a second reversal of the SnapMirror relationships, the production data is back at the primary site. This data is now protected in the same way that it was before the failover to the DR site—through SnapMirror and SnapVault backups.

Use of Qtrees in Site Recovery Manager environments

Qtrees are special directories that allow the application of file system quotas for NAS. ONTAP 9 allows the creation of qtrees, and qtrees can exist in volumes that are replicated with SnapMirror. However, SnapMirror does not allow replication of individual qtrees or qtree-level replication. All SnapMirror replication is at the volume level only. For this reason, NetApp does not recommend the use of qtrees with SRM.

Mixed FC and iSCSI environments

With the supported SAN protocols (FC, FCoE, and iSCSI), ONTAP 9 provides LUN services—that is, the ability to create and map LUNs to attached hosts. Because the cluster consists of multiple controllers, there are multiple logical paths that are managed by multipath I/O to any individual LUN. Asymmetric logical unit access (ALUA) is used on the hosts so that the optimized path to a LUN is selected and is made active for data transfer. If the optimized path to any LUN changes (for example, because the containing volume is moved), ONTAP 9 automatically recognizes and nondisruptively adjusts for this change. If the optimized path becomes unavailable, ONTAP can nondisruptively switch to any other available path.

VMware SRM and NetApp SRA support the use of the FC protocol at one site and the iSCSI protocol at the other site. It does not support having a mix of FC-attached datastores and iSCSI-attached datastores in the same ESXi host or in different hosts in the same cluster, however. This configuration is not supported with SRM because, during the SRM failover or test failover, SRM includes all FC and iSCSI initiators in the ESXi hosts in the request.

Best Practice
SRM and SRA support mixed FC and iSCSI protocols between the protected and recovery sites. However, each site should be configured with only one protocol, either FC or iSCSI, not both protocols at the same site. If a requirement exists to have both FC and iSCSI protocols configured at the same site, NetApp recommends that some hosts use iSCSI and other hosts use FC. NetApp also recommends in this case that SRM resource mappings be set up so that the VMs are configured to fail over into one group of hosts or the other.

Troubleshooting SRM when using vVols replication

The workflow within SRM is significantly different when using vVols replication from what is used with SRA and traditional datastores. For example, there is no array manager concept. As such, `discoverarrays` and `discoverdevices` commands are never seen.

When troubleshooting, it is beneficial to understand the new workflows, which are listed below:

1. queryReplicationPeer: Discovers the replication agreements between two fault domains.
2. queryFaultDomain: Discovers fault domain hierarchy.
3. queryReplicationGroup: Discovers the replication groups present in the source or target domains.
4. syncReplicationGroup: Synchronizes the data between source and target.
5. queryPointInTimeReplica: Discovers the point in time replicas on a target.
6. testFailoverReplicationGroupStart: Begins test failover.
7. testFailoverReplicationGroupStop: Ends test failover.
8. promoteReplicationGroup: Promotes a group currently in test to production.
9. prepareFailoverReplicationGroup: Prepares for a disaster recovery.
10. failoverReplicationGroup: Executes disaster recovery.
11. reverseReplicateGroup: Initiates reverse replication.
12. queryMatchingContainer: Finds containers (along with Hosts or Replication Groups) that might satisfy a provisioning request with a given policy.
13. queryResourceMetadata: Discovers the metadata of all resources from the VASA provider, the resource utilization can be returned as an answer to the queryMatchingContainer function.

The most common error seen when configuring vVols replication is a failure to discover the SnapMirror relationships. This occurs because the volumes and SnapMirror relationships are created outside of the purview of ONTAP Tools. Therefore, it is a best practice to always make sure your SnapMirror relationship is fully initialized and that you have run a rediscovery in ONTAP Tools at both sites before attempting to create a replicated vVols datastore.

Conclusion

VMware vCenter Site Recovery Manager is a disaster recovery offering that provides automated orchestration and nondisruptive testing of centralized recovery plans to simplify disaster recovery management for all virtualized applications.

By deploying Site Recovery Manager on NetApp ONTAP systems, you can dramatically lower the cost and complexity of disaster recovery. With high-performance, easy-to-manage, and scalable storage appliances and robust software offerings, NetApp offers flexible storage and data management solutions to support vSphere

environments.

The best practices and recommendations that are provided in this guide are not a one-size-fits-all solution. This document contains a collection of best practices and recommendations that provide guidelines to plan, deploy, and manage SRM DR plans. Consult with a local NetApp VMware expert when you plan and deploy VMware vCenter Site Recovery environments onto NetApp storage. NetApp VMware experts can quickly identify the needs and demands of any vSphere environment and can adjust the storage solution accordingly.

Additional Information

To learn more about the information that is described in this document, review the following documents and/or websites:

- TR-4597: VMware vSphere for ONTAP
  https://docs.netapp.com/us-en/netapp-solutions/hybrid-cloud/vsphere_ontap_ontap_for_vsphere.html
- TR-4400: VMware vSphere Virtual Volumes with ONTAP
  https://www.netapp.com/pdf.html?item=/media/13555-tr4400.pdf
- TR-4015 SnapMirror Configuration Best Practice Guide for ONTAP 9
  https://www.netapp.com/media/17229-tr4015.pdf?v=127202175503P
- RBAC User Creator for ONTAP
  https://mysupport.netapp.com/site/tools/tool-eula/rbac
- ONTAP tools for VMware vSphere Resources
  https://mysupport.netapp.com/site/products/all/details/otv/docsandkb-tab
- VMware Site Recovery Manager Documentation
  https://docs.vmware.com/en/Site-Recovery-Manager/index.html

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